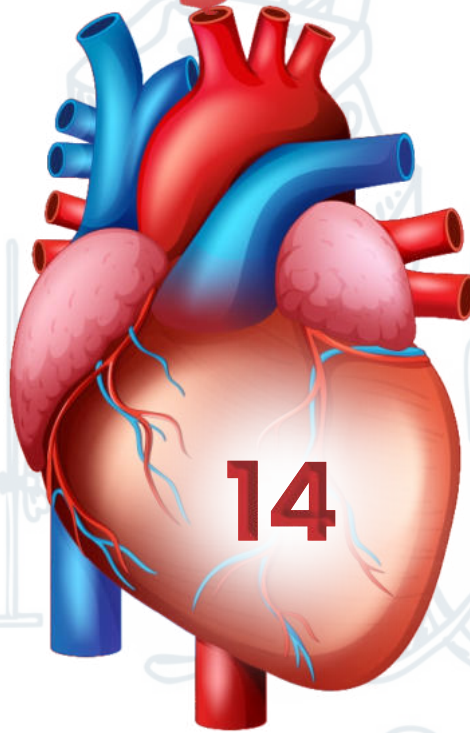
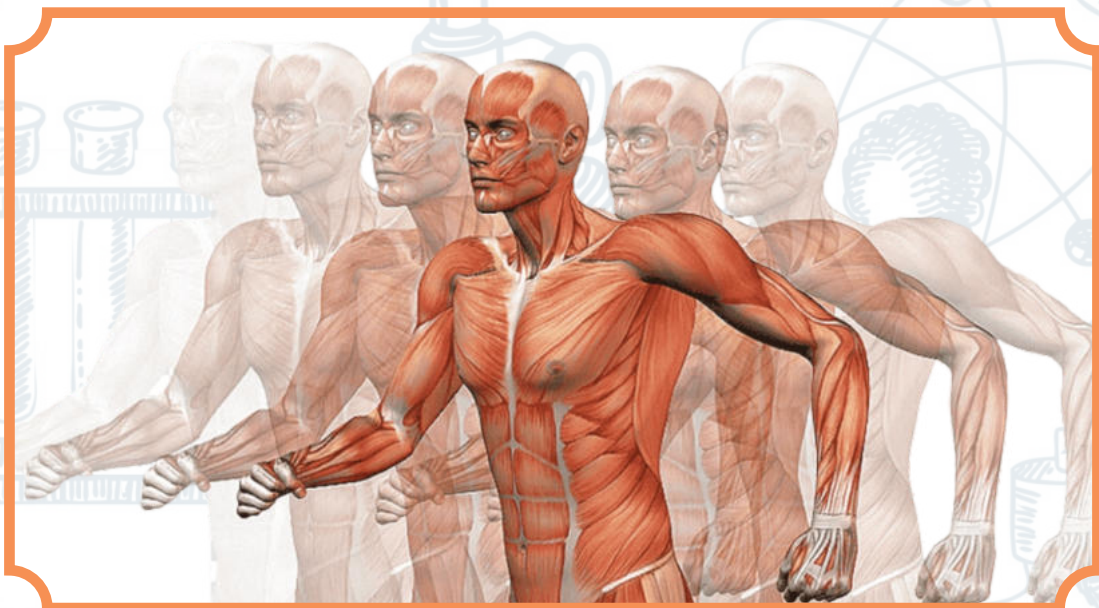


UNIT



GASEOUS EXCHANGE



Respiration and Gaseous Exchange

Respiration

Definition:

Respiration is the process through which cells gain energy by oxidizing food molecules (mainly glucose) to produce ATP.

Type of Respiration:

- **Aerobic Respiration:** Involves the oxidation of glucose using oxygen (O_2), resulting in energy (ATP), carbon dioxide (CO_2), and water as by-products.
- **Importance:** Oxygen is essential for this process, and it is continuously required by cells to generate ATP, especially in aerobic conditions.

Ventilation and Gaseous Exchange (14.1.1)

Ventilation:

The process of moving air in and out of the lungs is known as ventilation. This helps provide oxygen (O_2) to the body and remove carbon dioxide (CO_2).

Gaseous Exchange:

The lungs facilitate the exchange of gases between the air and the blood. This occurs in the alveoli, which are tiny, sac-like structures where the exchange of O_2 and CO_2 takes place.

Anatomy of the Lungs:

- **Airways:** Lungs have a branching system of airways that end in microscopic sac-like structures known as alveoli.
- **Respiratory Bronchioles:** The smallest airways that lead to the alveolar ducts and sacs.
- **Alveolar Sacs:** Clusters of alveoli, the primary sites for gas exchange.
- **Alveoli:** Tiny sacs lined with a thin film of moisture that facilitate the exchange of gases with capillaries.

Mechanism of Gaseous Exchange:

- **Alveolar Air:** Oxygen from the air in the alveoli diffuses into the capillaries (small blood vessels) surrounding the alveoli.
- **Capillary Blood:** Carbon dioxide from the blood in the capillaries diffuses into the alveolar air to be exhaled.

Ventilation and Blood Flow:

- Effective Gaseous Exchange requires proper ventilation of the alveoli and sufficient blood flow through the surrounding capillaries.

Respiratory Surface

Definition:

- The respiratory surface is the area where the exchange of gases (oxygen and carbon dioxide) occurs between an organism and its environment.

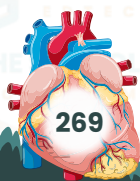
Examples of Respiratory Surfaces:

- **Earthworms:** Use their skin as the respiratory surface.
- **Humans:** Use the alveolar membrane in the lungs for gaseous exchange.

Characteristics of a Respiratory Surface (Alveoli)

1. Moist:

- The respiratory surface must be moist to allow gases to dissolve and diffuse through the membrane (e.g., oxygen and carbon dioxide).



1. Permeable:

- The surface should be permeable to respiratory gases (e.g., oxygen and carbon dioxide) to allow them to pass through easily.

2. Thin:

- The respiratory surface should be thin to reduce the diffusion distance, allowing faster exchange of gases.

3. Large Surface Area:

- The respiratory surface should have a large surface area relative to the volume of the organism. In humans, the alveoli provide a large surface area to facilitate efficient gas exchange. The combined surface area of alveoli in the lungs is several times larger than the total body surface.

Main Components of the Human Respiratory System

The human respiratory system consists of a series of organs and structures that work together to bring air from the environment into the lungs and remove carbon dioxide from the body.

1. Nose (External Nostril):

- Air enters the respiratory system through the nose, where it is filtered, warmed, and moistened.

2. Pharynx (Throat):

- The pharynx serves as a passageway for air moving from the nasal cavity to the larynx.
- It also connects to the esophagus, which carries food.

3. Larynx (Voice Box):

- Located at the top of the trachea, the larynx houses the vocal cords.
- It serves as a passage for air and helps protect the trachea against food aspiration by closing during swallowing.

4. Trachea (Windpipe):

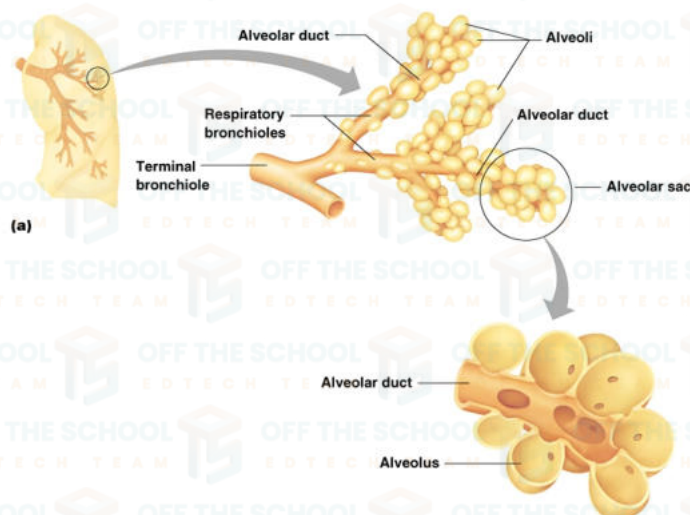
- The trachea is a rigid tube that extends from the larynx to the bronchi.
- It conducts air to the lungs and is supported by cartilage rings to prevent collapsing.

5. Bronchi (Main Branches):

- The trachea divides into two bronchi (right and left), one leading to each lung.
- The bronchi further divide into smaller bronchioles within the lungs.

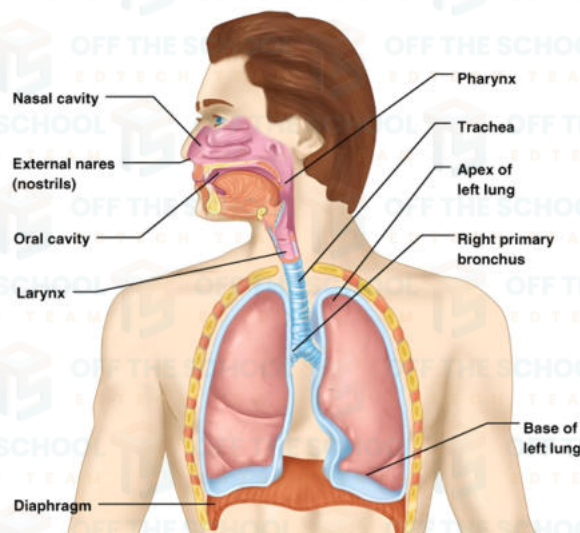
6. Bronchioles:

- These are the smaller branches of the bronchi that lead air deeper into the lungs.
- The bronchioles end in the alveolar sacs, which contain the alveoli for gas exchange.



7. Lungs:

- The lungs are paired, spongy, and vascularized organs located within the thoracic (chest) cavity.
- The lungs are responsible for taking in oxygen and releasing carbon dioxide.
- The right lung has three lobes, while the left lung has two, to accommodate the heart.
- Alveoli are located in the lungs and are the site where oxygen from the air is exchanged with carbon dioxide from the blood.



Anatomy of the Human Respiratory System

1. Upper and Lower Respiratory Air Passages

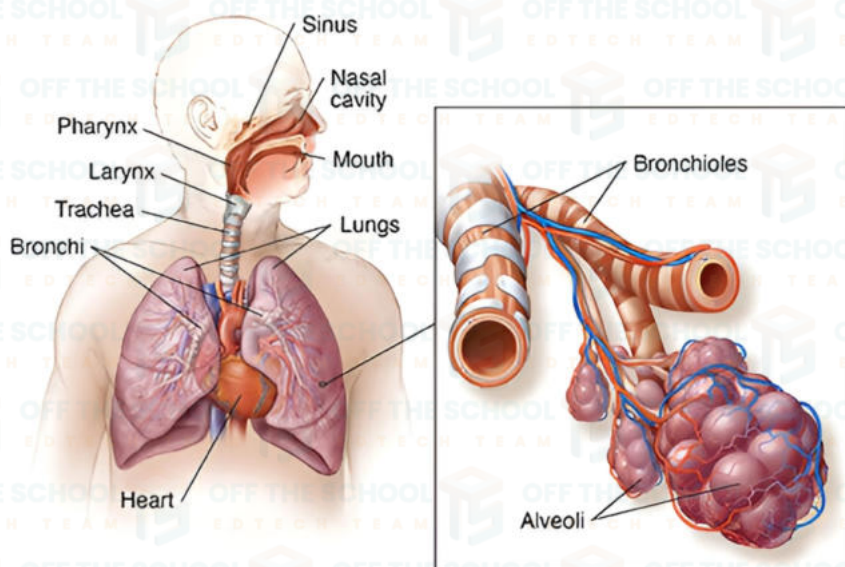
The human respiratory system is divided into upper and lower respiratory air passages:

Upper Respiratory Passage:

- **Nose:** Air enters the respiratory system through external nostrils or nares. The nose is divided into two chambers by a bony partition and a cartilaginous nasal septum.
- **Nasal Cavity:** Located inside the nose, it filters and humidifies the air. Ciliated cells trap dust particles to clean the air.
- **Pharynx:** A muscular tube about 13 cm long, it serves as a common pathway for both the respiratory and digestive systems. It has three regions:
- **Nasopharynx:** Air passes through the internal nares into this region.
- **Oropharynx:** Behind the mouth, it connects to the larynx.
- **Laryngopharynx:** Directs air towards the larynx.

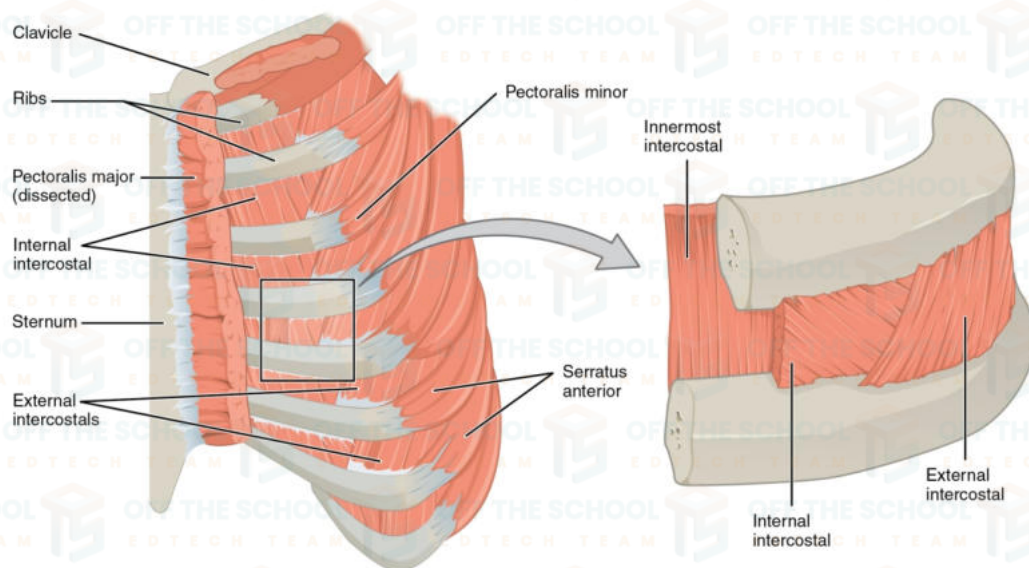
Lower Respiratory Passage:

- **Larynx (Voice Box):** Located at the entrance of the trachea, the larynx houses the vocal cords that produce sound when air is expelled.
- **Trachea (Windpipe):** A fibro-cartilaginous tube about 10-11 cm long with 16-20 "C" shaped cartilage rings to maintain rigidity and prevent collapse.
- **Bronchi:** The trachea bifurcates into the left and right bronchi. The right bronchus is shorter, wider, and straighter than the left. These bronchi branch into smaller bronchioles.
- **Bronchioles:** Smaller branches of the bronchi that end at the alveoli.
- **Alveoli:** Tiny air sacs where gas exchange (oxygen and carbon dioxide) takes place between the air and blood.



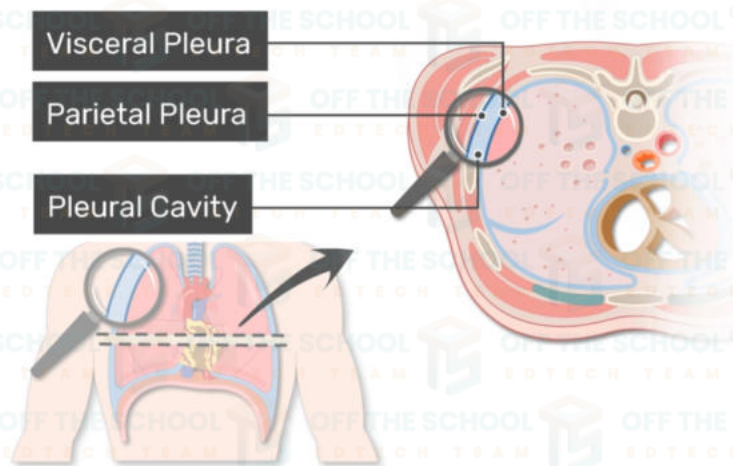
2. Thoracic Cavity and Rib Cage

- **Rib Cage:** Consists of 12 pairs of ribs that protect the lungs and heart.
- **Thoracic Cavity:** The chest cavity that houses the lungs and heart, surrounded by the rib cage. It is separated from the abdominal cavity by the diaphragm.
- **Intercostal Muscles:** Located between the ribs, they assist in breathing by expanding and contracting the rib cage.



3. Pleura and Pleural Cavity

- **Pleura Membranes:** Two layers of tissue surrounding the lungs:
- **Visceral Pleura:** Wraps around each lung.
- **Parietal Pleura:** Lines the thoracic cavity.
- **Pleural Cavity:** The space between the visceral and parietal pleura filled with serous fluid. This fluid acts as a lubricant to allow the lungs to expand and contract smoothly during breathing.



Lungs and Respiratory Functions

1. Structure of the Lungs

- **Lungs:** The primary respiratory organs responsible for gas exchange. The left and right lungs differ slightly in structure:
- **Left Lung:** Smaller than the right lung, as it is slightly displaced by the heart, which occupies more space on the left side of the thorax.
- **Right Lung:** Larger than the left lung and consists of three lobes.

Lobes of the Lungs:

Right Lung:

- Superior Lobe
- Middle Lobe
- Inferior Lobe

Left Lung:

- Superior Lobe
- Inferior Lobe

Fissures:

- The right lung has a horizontal and oblique fissure separating its lobes.
- The left lung has an oblique fissure.

2. Major Functions of the Respiratory System

The respiratory system performs several vital functions:

Oxygen Supply:

- Delivers a continuous supply of oxygen to all tissues in the body.

Removal of Carbon Dioxide:

- Facilitates the removal of carbon dioxide (CO₂), a by-product of cellular respiration.

Gas Exchange:

- Regulates the mechanism of gas exchange between the internal and external environments via the respiratory surfaces (alveoli).

Humidification of Air:

- The respiratory system humidifies and warms the air that is inhaled, ensuring that it is at the optimal temperature and moisture level for gas exchange.

These functions help maintain homeostasis by ensuring proper oxygen levels in the blood, eliminating carbon dioxide, and preparing the air for efficient gas exchange in the lungs.

Ventilation (Breathing)

1. Definition of Ventilation

- Ventilation is the process of air moving in and out of the lungs.
- It involves two phases: Inhalation and Exhalation, both of which are essential for the breathing cycle and occur continuously throughout life.

2. Phases of Breathing:

- **Inhalation (Inspiration):** The process of drawing air into the lungs.
- **Exhalation (Expiration):** The process of expelling air from the lungs.

3. Mechanism of Ventilation

- The movement of air in and out of the lungs is driven by pressure changes within the lungs compared to the external atmospheric pressure.
- **Negative Pressure Breathing:** During both inhalation and exhalation, the pressure inside the thoracic cavity changes in relation to external atmospheric pressure, facilitating airflow.

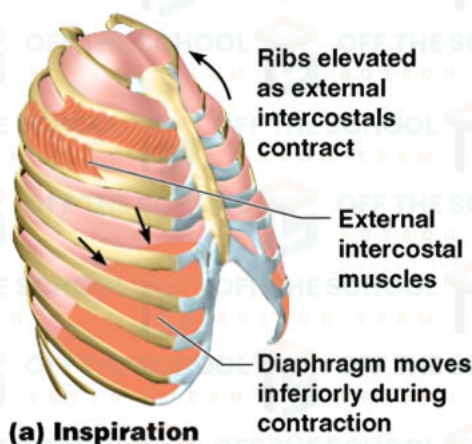
4. Inspiration (Inhalation)

- **Active Process:** Requires energy.

Mechanism:

- **Thoracic Cavity Expansion:** The diaphragm contracts and flattens, and the external intercostal muscles contract, lifting the ribs and sternum.
- **Pressure Decreases:** As the volume of the thoracic cavity increases, the pressure inside the lungs drops below atmospheric pressure.
- **Air Rushes In:** The lower pressure inside the lungs causes air to flow into the alveoli from the outside environment.

Changes in anterior-posterior and superior-inferior dimensions



(a) Inspiration

Changes in lateral dimensions

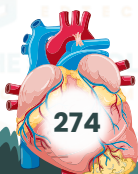


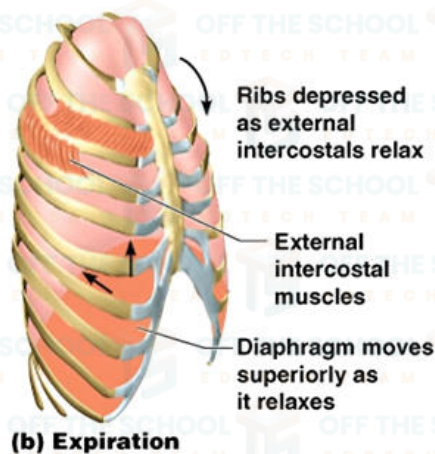
5. Expiration (Exhalation)

- **Passive Process:** Does not require energy.

Mechanism:

- **Thoracic Cavity Compression:** The diaphragm relaxes (returning to its dome shape), and the external intercostal muscles relax while the internal intercostal muscles contract.
- **Pressure Increases:** The volume of the thoracic cavity decreases, increasing the pressure inside the lungs above atmospheric pressure.
- **Air is Expelled:** Due to the higher internal pressure, air is forced out of the lungs through the respiratory passages.





6. Breathing Rate

- The breathing rate refers to how often a person breathes in a minute. It may vary depending on factors such as activity level (e.g., increases during exercise), health, and age.

Lung Volume and Capacity

1. Definition of Lung Volume

- Lung volumes refer to the amounts of air that the lungs can hold at various stages of the breathing cycle.
- The total lung capacity for an average human adult is around 6 liters of air.

2. Measurement of Lung Volume

- Spirometry is the special instrument used to measure lung volumes.
- The values of lung volumes may vary based on factors such as:
 - Race
 - Gender
 - Age
 - Body composition
 - Respiratory diseases

3. Types of Lung Volumes

1. Tidal Volume (TV):

- The volume of air taken in or expelled with each normal breath.
- Approx. 450–500 milliliters in normal breathing.

2. Vital Capacity (VC):

- The maximum volume of air that can be inspired and expired during deep breathing.
- Approx. 5000 milliliters.

3. Expiratory Reserve Volume (ERV):

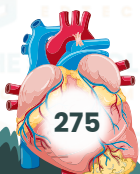
- The amount of air that can be forcefully exhaled after normal exhalation.
- Approx. 1200–1500 milliliters.

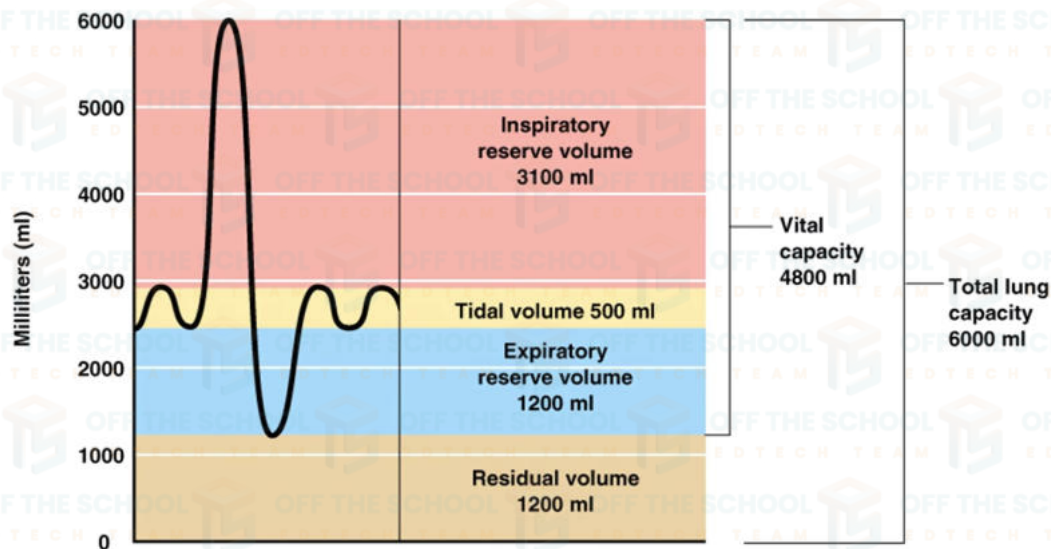
4. Inspiratory Reserve Volume (IRV):

- The additional volume of air that can be inhaled after normal inhalation.
- Approx. 2000 milliliters.

5. Residual Volume (RV):

- The volume of air that remains in the lungs after full exhalation.
- Approx. 1000 milliliters.
- Prevents the lungs from collapsing.





4. Factors Affecting Lung Volume

- Lung volumes can change due to various factors:
- Age:** Lung volume tends to decrease with age.
- Respiratory diseases:** Conditions such as asthma, COPD, etc., can alter lung capacity.
- Body composition and physical fitness:** Well-conditioned individuals may have better lung capacities.

5. Controlling Mechanism of Breathing

1. Voluntary Control:

- Breathing can be consciously controlled, such as when holding the breath or breathing more deeply or rapidly.

2. Involuntary Control:

- Normal, automatic breathing is regulated by the respiratory centers in the brain.
- Chemoreceptors (located in the aorta and carotid arteries) monitor the levels of carbon dioxide (CO_2) and hydrogen ions (H^+) in the blood.
- Medulla Oblongata: Detects changes in CO_2 and H^+ concentrations in the cerebrospinal fluid and sends signals to the diaphragm and intercostal muscles to adjust the rate of breathing accordingly.

6. Regulation of Breathing Rate

- High CO_2 and H^+ levels in the blood stimulate an increase in the rate and depth of breathing.
- The process is automatic and controlled by the interaction of the respiratory and cardiovascular systems.

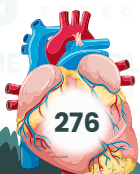
Transportation of Gases in Humans

1. General Overview

- The exchange of gases (oxygen and carbon dioxide) is a vital process in respiration.
- Oxygen (O_2) and carbon dioxide (CO_2) are transported in the human body by blood circulation.
- These gases are exchanged in the alveolar membrane via simple diffusion.

2. Oxygenation of Blood:

- The oxygenated blood becomes bright red due to the oxygen binding to hemoglobin in red blood cells (RBCs).



- About 97% of oxygen is carried by RBCs bound to hemoglobin, while the remaining 3% is dissolved in blood plasma.

3. Transport to Body Cells:

- Oxygenated blood is transported from the lungs to body cells.
- In the tissues, oxygen detaches from hemoglobin and diffuses into the cells.

4. Cellular Respiration:

- Inside the cells, oxygen helps break down glucose to release energy, producing CO₂, water, and energy as by-products.

3. Transport of Carbon Dioxide (CO₂)

1. Formation of CO₂:

- Carbon dioxide is produced as a by-product of cellular respiration.
- It diffuses out from the tissues into the blood.

2. Transport to the Lungs:

- The deoxygenated blood, now carrying CO₂, is transported back to the lungs for exhalation.

3. Modes of Transport of CO₂:

As bicarbonate ions (HCO₃⁻):

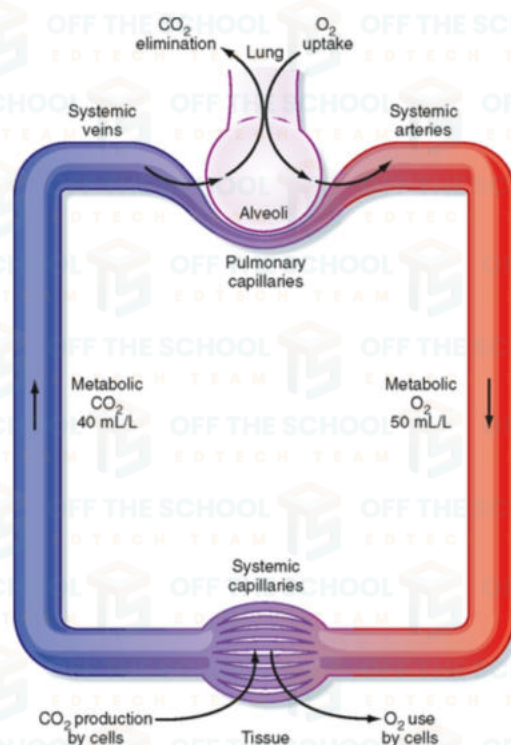
- About 70% of CO₂ is transported in the form of bicarbonate (HCO₃⁻) in RBCs, where the enzyme carbonic anhydrase converts CO₂ and water into carbonic acid (H₂CO₃), which dissociates into hydrogen ions (H⁺) and bicarbonate ions (HCO₃⁻).
- Bicarbonate ions combine with sodium (Na⁺) or potassium (K⁺) to form sodium bicarbonate (NaHCO₃) or potassium bicarbonate (KHCO₃), respectively.

As carbaminohemoglobin:

- About 20% of CO₂ binds with hemoglobin to form carbaminohemoglobin.
- This compound dissociates in the lungs, releasing CO₂, which is then exhaled.

As dissolved gas:

- Approximately 10% of CO₂ is dissolved directly in blood plasma.



4. Exhalation of CO₂

- In the lungs, the partial pressure of CO₂ is lower compared to the tissues, causing CO₂ to dissociate from carbaminohemoglobin and diffuse into the alveoli.
- The dissolved CO₂ in the plasma and bicarbonate ions also move into the alveoli to be exhaled.

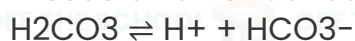
Chemical Reactions Involved in CO₂ Transport:

Formation of Carbonic Acid:



(Carbon dioxide + Water = Carbonic acid)

Dissociation of Carbonic Acid:

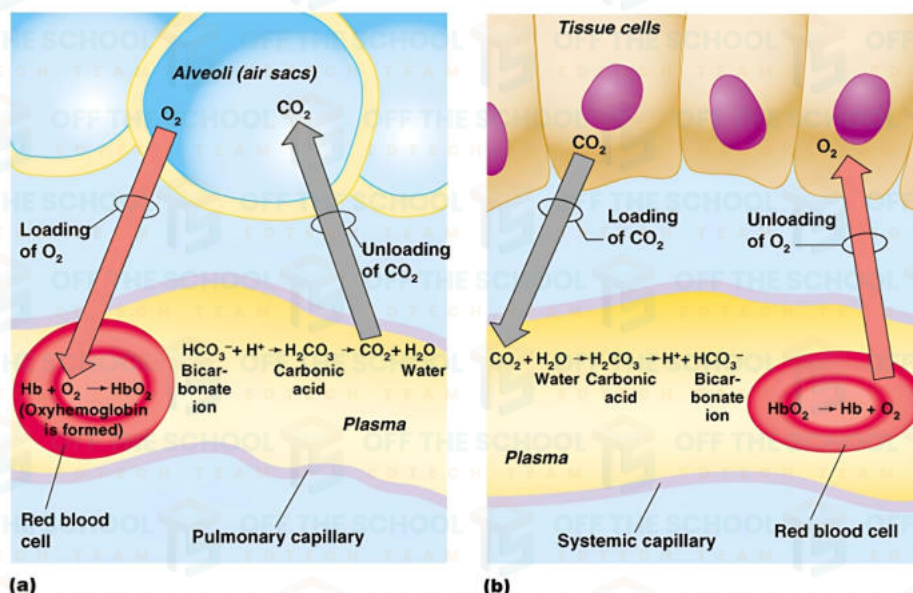


(Carbonic acid dissociates into hydrogen ions and bicarbonate ions)

Carbaminohemoglobin Formation:



(Carbon dioxide binds with hemoglobin to form carbaminohemoglobin)



Role of Respiratory Pigments

1. Definition

- Respiratory pigments are molecules that enhance the oxygen-carrying capacity of blood and tissues.
- They bind with oxygen to facilitate its transport through the bloodstream and tissues.

2. Hemoglobin (Hb)

1. Composition:

- Hemoglobin is a conjugated protein found in red blood cells (RBCs) of vertebrates, including humans.
- It consists of four protein subunits, each containing an iron-containing heme group.
- Each hemoglobin molecule can bind with four molecules of oxygen, making it highly efficient for oxygen transport.

2. Oxygen Transport:

- When hemoglobin binds with oxygen in the lungs, it forms oxyhemoglobin, a temporary compound.

- Oxyhemoglobin carries oxygen from the lungs to body tissues.
- In the tissues, oxygen dissociates from oxyhemoglobin, allowing it to diffuse into the cells.

3. Carbon Dioxide Transport:

- Hemoglobin also helps transport carbon dioxide (CO_2) back from the tissues to the lungs for exhalation.

Formation and Dissociation of Oxyhemoglobin:

- Oxygen binds to the iron in the heme group to form oxyhemoglobin.
- Oxygen is released from oxyhemoglobin in the tissues, and hemoglobin returns to its deoxygenated form.

1. Myoglobin

1. Composition:

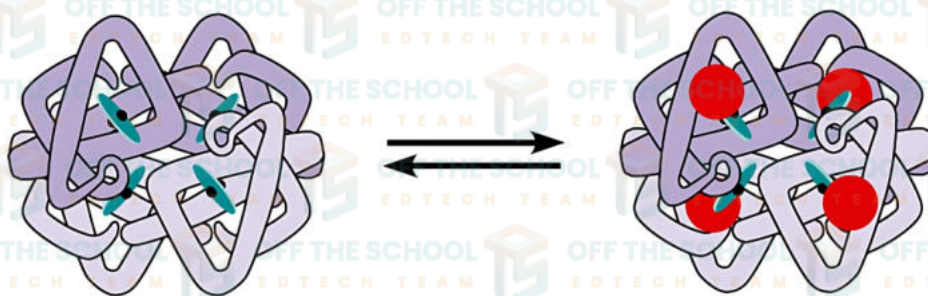
- Myoglobin is a smaller, single-chain globular protein found in muscles.
- Like hemoglobin, myoglobin also contains a heme group (iron-containing organic compound).

2. Oxygen Binding:

- Myoglobin binds oxygen more tightly than hemoglobin and can hold only one molecule of oxygen per heme group.
- Myoglobin is responsible for storing oxygen in muscles and releasing it when needed, particularly during muscle activity.

Key Functions of Respiratory Pigments:

- **Hemoglobin:** Efficient oxygen transport from lungs to tissues and carbon dioxide transport from tissues to lungs.
- **Myoglobin:** Oxygen storage and release in muscles, ensuring an adequate oxygen supply during physical activity.



Sinuses

Definition:

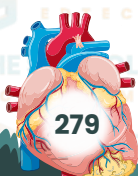
- Sinuses are hollow cavities or air spaces found within the bones of the skull. They are connected to the nasal cavities and play a role in air passage, humidity regulation, and sound resonance.

Types of Sinuses:

There are four pairs of sinuses in humans, each located in different parts of the skull:

1. Frontal Sinus:

- **Location:** In the forehead region, above the eyes.
- **Function:** Helps in lightening the weight of the skull and contributing to voice resonance.



2. Maxillary Sinus:

- **Location:** Behind the cheekbones (on each side of the nose).
- **Function:** Helps in humidifying and warming the air we breathe, and also contributes to voice resonance.

3. Ethmoid Sinus:

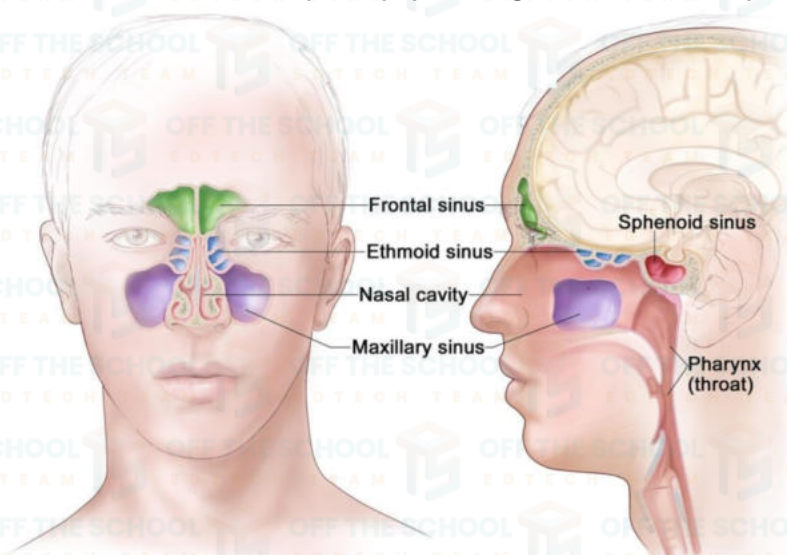
- **Location:** Between the eyes, behind the bridge of the nose.
- **Function:** Helps in air filtration and humidification, and also aids in voice resonance.

4. Sphenoid Sinus:

- **Location:** Deep within the skull, behind the ethmoid sinus.
- **Function:** Helps in reducing skull weight and aiding in voice resonance.

Key Functions of Sinuses:

1. **Lightening the Skull:** Sinuses reduce the overall weight of the skull.
2. **Humidification and Warming:** They help in humidifying and warming the air before it reaches the lungs.
3. **Resonance of Voice:** Sinuses contribute to the quality of the human voice by acting as resonating chambers.
4. **Protection and Filtration:** Sinuses help trap pathogens and dust particles from the air.



Respiratory Disorders

Respiratory disorders are classified into two major groups based on the part of the respiratory system they affect:

1. Upper Respiratory Tract Infections

2. Lower Respiratory Tract Infections

1. Upper Respiratory Tract Infections

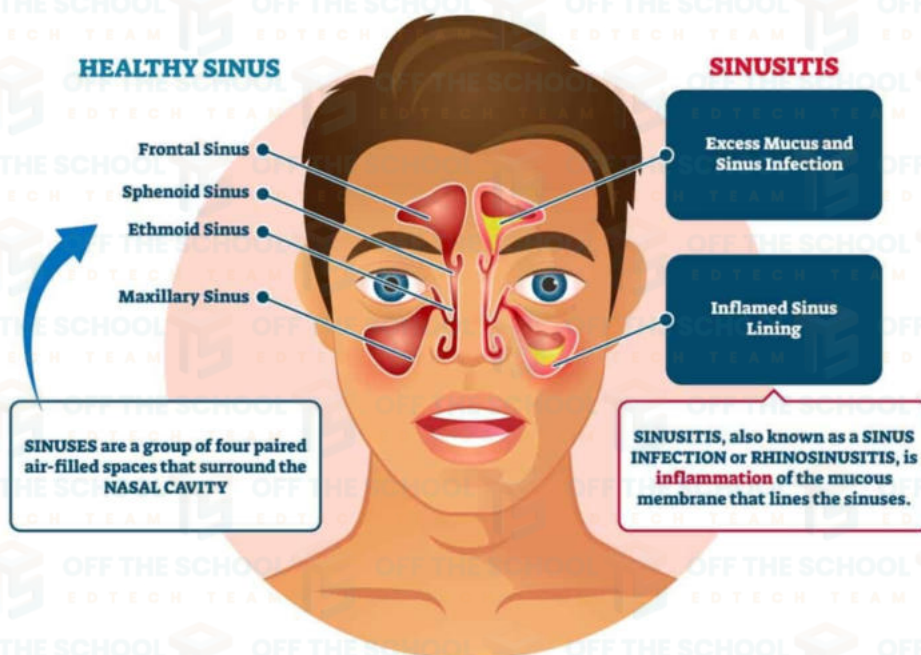
i) Sinusitis

• **Definition:** Inflammation of the sinuses, which are normally filled with air. Sinusitis occurs when the sinuses become filled with fluid and may harbor pathogens.

Causes:

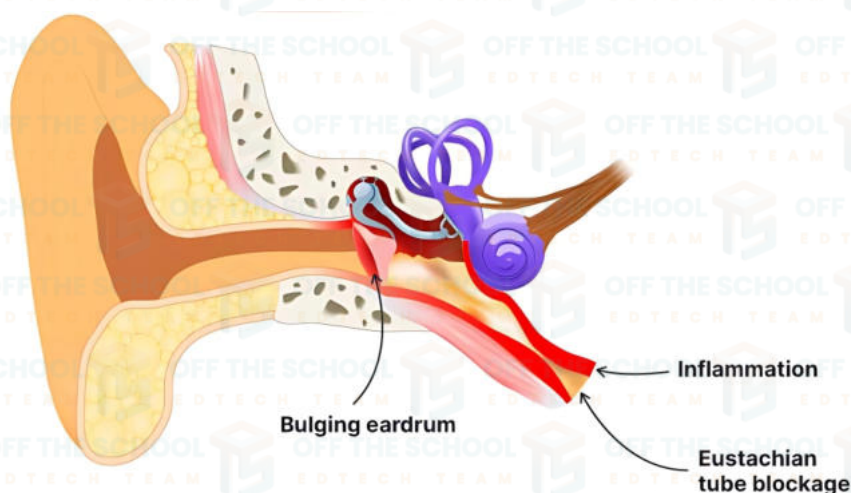
- Common cold
- Allergic rhinitis (swelling in the lining of the nose)
- Growths (polyps) in the nasal passage
- Dislocation of the nasal septum

- Headache
- Fever
- Nasal congestion (stuffy nose)
- Cough
- Tooth pain
- Ear pain
- **Treatment:** Consult a physician for proper diagnosis and use saline nasal sprays or other prescribed medications.



ii) Otitis Media

- **Definition:** Inflammation or infection in the middle ear, often caused by failure of the Eustachian tube to regulate pressure between the ear and throat.
- **Symptoms:**
 - Ear pain
 - Fever
 - Unbalanced sensation
 - Hearing problems
 - Irritability (especially in children)
- **Treatment:** Treatment may include ear drops or antibiotics as prescribed by a doctor.



2. Lower Respiratory Infections

i) Pneumonia

- **Definition:** A lower respiratory infection affecting the lungs, caused by bacteria, viruses, or fungi. The infection fills the air sacs of the lungs with fluid and secretions.
- **Types:**
 - Lobar Pneumonia: Affects the lobes of the lungs.
 - Bronchial Pneumonia (Bronchopneumonia): Causes patches of infection in both lungs.
- **Symptoms:**
 - Headache
 - Bluish lips and fingernails (due to lack of oxygen)
 - Fever with cough
 - Yellow, green, or bloody mucus with cough
 - Confusion
 - Heavy sweating
 - Loss of appetite
 - Rapid breathing and shortness of breath
- **Treatment:**
 - **Bacterial pneumonia:** Treated with oral antibiotics.
 - **Viral pneumonia:** No antibiotics; symptom management is used (e.g., hot drinks, humidification, steamy baths).
 - **Mycoplasma pneumonia:** May require antibiotics and specific treatment for acute cough with mucus.

